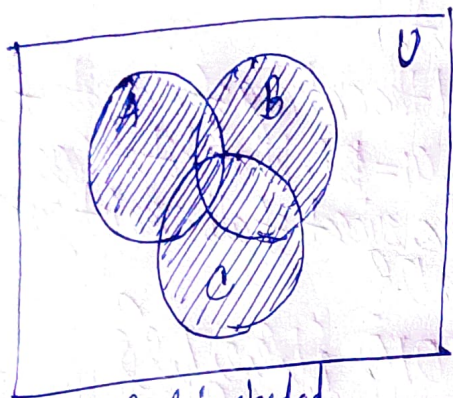
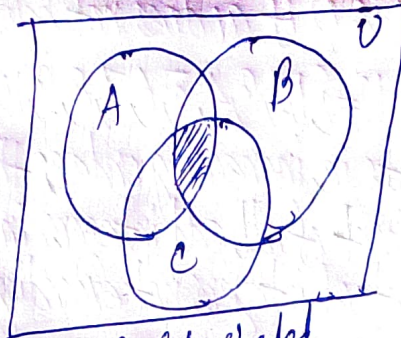




(a) $A \cap B \cap C$ is shaded



(b) $A \cap B \cap C$ is shaded

Example 15:

Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$ and $C = \{0, 3, 6, 9\}$. What are

$A \cap B \cap C$ and $A \cap B \cap C$?

Solution:

$$A \cap B \cap C = \{0, 1, 2, 3, 4, 6, 8, 9\}$$

$$A \cap B \cap C = \{0, 3\}$$

Definition 6: The **union** of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

We use the notation $A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$ to denote the union of the sets $A_1, A_2, \dots, A_n$.

Definition 7: The **intersection** of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

We use the notation $A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$ to denote the intersection of the sets $A_1, A_2, \dots, A_n$.

2.2.4 Computer Representation of Sets

We will present a method for storing elements using an arbitrary ordering of the elements of the universal set. This method of representing sets makes computing combinations of sets easy.

Assume that the universal set U is finite (and of reasonable size so that the number of elements of U is not larger than the memory size of the computer being used.) First, specify an arbitrary ordering of the elements of U , for instance $a_1, a_2, \dots, a_n$. Represent a subset A of U with the bit string of length n , where the i th bit in this string is 1 if a_i